instacart

Case Study

Instacart Agent Readiness – Product



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The idea behind my answers

Agents won't fail at Instacart because they can't read a catalog. They'll fail at the small moments a person decides without thinking: **which oat milk** when yours is sold out, whether **"2" means bags or kilos**, whether **the 5th order this month was really wanted**. So I find those moments, make each decision **readable by a machine**, and let the agent's freedom grow **only as fast as the evidence**.

1 Why groceries first

The job **repeats weekly**, the customer's **rules are already on file**, and a mistake costs a **line-item refund**.

Loyal users order 3.9×/month · 3 saved replacement options · #1 category people would let AI buy (28%)

2 The flashy board demo

I build a demo, not the throwaway. It runs on the real substitution feature; scripted parts capped at **5 engineer-days**.

ChatGPT in-chat checkout was pulled after ~6 months; Walmart saw it convert about 3× worse

3 Checkout or substitutions

Substitutions. Checkout is a permissions problem others are already solving; substitution is a **judgment call on a live clock**.

Instacart's public agent tools: 2, none for replacements · scoring 4.8 vs 1.8

4 "quantity: 2" read two ways

Our contract is the bug, not the agents. Required unit + expected-total check + a **read-back** of what we understood.

Instacart itself retired its bare quantity field for a measurements list in 2026

5 Orders per user per month

4 on its own, max 1 per rolling 7 days; the 5th needs one tap. Limits rise only with evidence.

Loyal users average 3.9/month · 300 clean orders ⇒ unwanted-order rate under 1% (95% confidence)

6 "Should we be worried?"

Honestly, a little: enough to act, not to stop. Name the failure modes, sign 3 tripwires, keep an assumptions log.

Gartner: 40%+ of agent projects cancelled by 2027 · Instacart's own $60M FTC consent case

90.3M orders last quarter, +9% YoY · **$115** average order value · **1.6B** orders ever placed · **2B+** catalog items, 2,200 retailers · **3** AI platforms: ChatGPT, Claude, Gemini · **1** own agent: Clementine

Figures: Instacart Q2 2026 results [5]; Clementine launch coverage [4]; ChatGPT, Claude and Gemini integrations [1][2][3]. Full sources on slide 8.

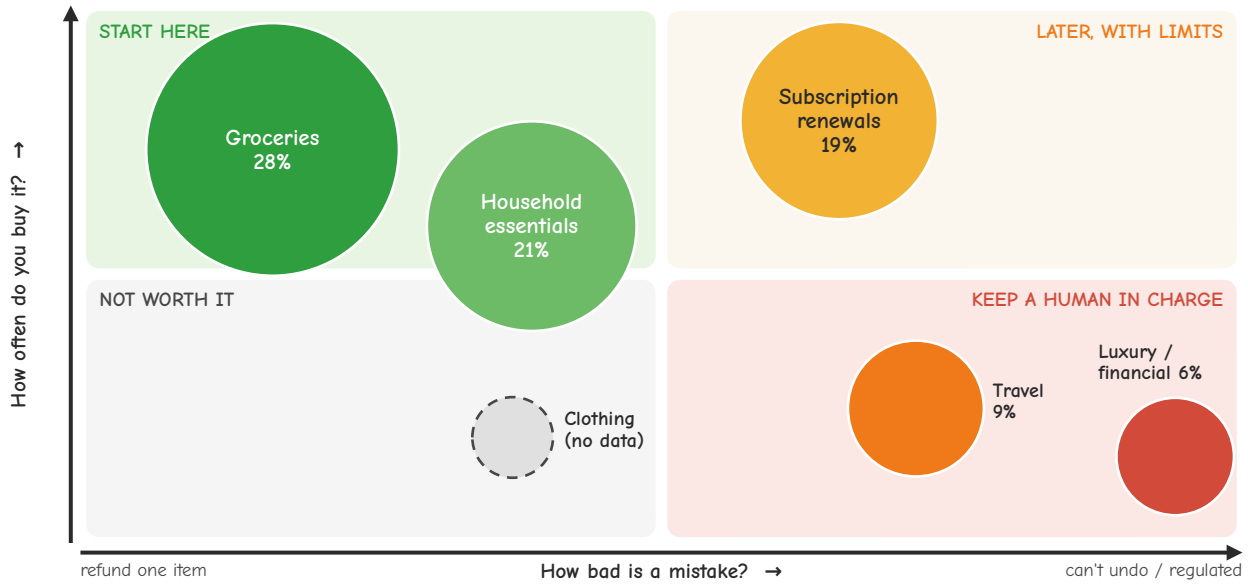
QUESTION 1 OF 6

Why start with groceries?

The question: Make the case for why groceries are a good place to start.

My answer Groceries are the one big category where the job repeats every week, the customer's rules are already written down, and a wrong call costs a line-item refund, not a lost holiday.

Where should an AI shopper start? (bubble size = how many people would let AI buy it)



Bubble = % of US adults comfortable letting AI buy in that category [17]. ACI found the same order: groceries 41%, household 31% [18]. Where each category sits on the axes is my own judgment.

Why grocery is hard, and why that's the point

About **two-thirds of Instacart orders have meat or produce** [5], and stock changes constantly (**10M+ inventory signals a day** [4]). The agent's value isn't typing a list; it's **handling the stock-out the way you would**. That's why Q3 picks substitutions. **My wedge:** "refill my usual basket within my rules". **Wedge metric:** % of recurring baskets delivered with zero edits and zero issues.

Framework I used: 4R check: Repeat · Rulebook · Reversible · Record

REPEAT

3.9×

orders / month

Instacart's monthly customers go from **2.1 orders/month in year 1 to 3.9 by year 6** [7]. US shoppers make **1.6 grocery trips a week** [23]. Instacart's own agent already handles "**order my usuals**" [4].

RULEBOOK

3

options on file

Per item, customers already pick **up to 3 backups, best match, or refund**, and approved replacements are **saved for next time**, with tags like same brand or gluten-free [10]. The agent inherits a brief; it doesn't invent one.

REVERSIBLE

95%

happy with a replacement

Orders with a replacement get **95% satisfaction**, and replacements per order are at their **lowest since 2022** [6]. A wrong item is a line-item refund, not a non-refundable fare.

RECORD

1.6B

orders ever

2B+ catalog items, 2,200 retailers [4]. Order quality is already measured and has improved **16 quarters in a row** [5], so agent orders can be judged against **normal orders from day one**.

Why now

Gemini can now **operate Android apps itself** [24], and a US appeals court ruled a **user-directed agent's access counts as the user's** [16]. Agents are coming anyway: better a front door than scraping.

Honest counterweight

Only **22% of AI shoppers trust agents to decide purchases** [19]. So launch is **opt-in and recurring-basket only**, with drafts before autonomy (Q5).

[4] Clementine launch · [5] Instacart Q2 2026 results · [6] Q2 2026 earnings call · [7] Instacart S-1 / Q3 2023 call · [10] Instacart Help Center · [16] Ninth Circuit, Amazon v. Perplexity · [17] Supermarket News · [18] ACI Worldwide · [19] FoodNavigator · [23] FMI Trends · [24] 9to5Google

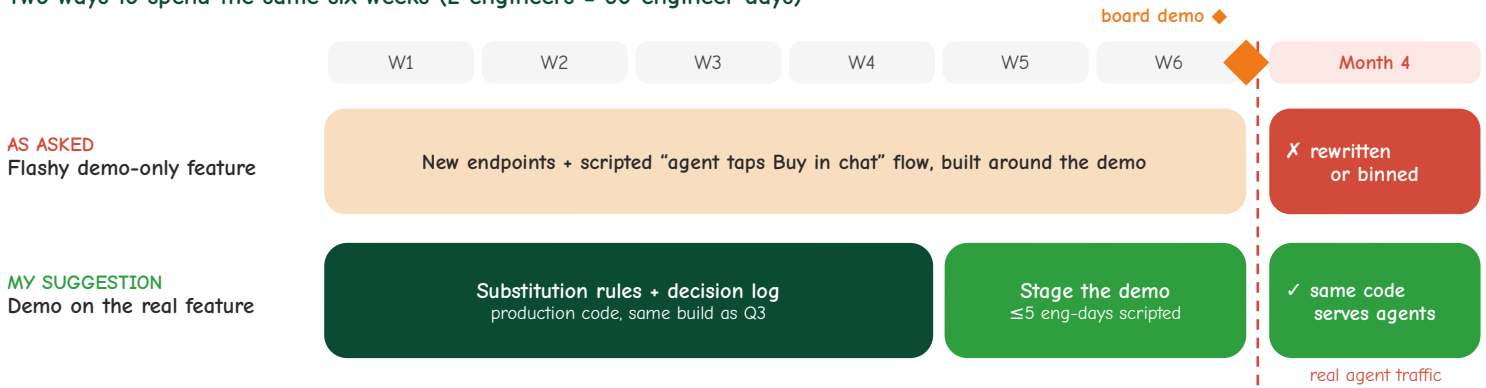
QUESTION 2 OF 6

Should I build the flashy board demo?

The question: The product team wants a flashier feature for a board demo in 6 weeks; I think real agent traffic will get it thrown out in 3 months. Build it, or push back and delay?

My answer — **Neither as asked, nor a delay. I build a board demo on top of the real substitution feature, so what the board sees in week 6 is still running in month 4. Scripted parts are capped at 5 engineer-days.**

Two ways to spend the same six weeks (2 engineers = 60 engineer-days)



Salvage test: every demo piece goes into one bin before anyone builds it

KEEP · real, stays

~48 eng-days · 80%

- Substitution rules API
- Decision log: saw → rule → chose
- Timeout + refund fallback
- Agent tools (MCP)

WRAP · scripted, calls real API

~7 eng-days · 12%

- Fake stock-out in the inventory feed
- Demo family + usual basket
- Presenter screen on real responses

THROW · pure theatre

≤5 eng-days · 8% (cap)

- Stage animation and narration
- Hard-coded "wow" timing
- Anything that changes the real API: refused

On stage: the stock-out save (everything real except the stock-out)



Framework I used: One-way vs two-way doors + tracer bullet

- **Two-way doors** (screens, narration, demo data): easy to undo, so build fast and time-box.
- **One-way doors** (real API contract, data model, public agent tools): **no changes for the demo**. This is the only line I'd push back on.
- **Tracer bullet:** a thin but **real** end-to-end path the team keeps building on, not a mock we rebuild later.

Evidence: the flashiest step hasn't lasted

- **ChatGPT in-chat checkout:** Instacart launched it first [1]; OpenAI **retired it about 6 months later** and moved checkout back to merchants [12].
- **Walmart:** in-chat purchases converted at **a third of the rate** of its own site; it called the experience "unsatisfying" [11].
- **Gartner:** over **40% of agent projects will be cancelled by 2027**, for cost, unclear value or weak risk controls [13].

How I'd say it, and if they still insist

"Let's give the board something still standing in month four. Tell me the moment you want on stage; I'll build it on the real feature and say what's scripted." If they still insist: they sign the salvage test, I log my concern (disagree and commit), and I escalate **only if it changes the real API**.

[1] Instacart × ChatGPT launch · [11] Search Engine Land, Walmart ChatGPT checkout · [12] CNBC, OpenAI revamps ChatGPT shopping · [13] Gartner. The effort split and 5-day cap are my proposals to negotiate with the team.

QUESTION 3 OF 6

Checkout or substitutions: which one first?

The question: You have 2 engineers and 4 weeks. Make the checkout API agent-readable, or the substitution-preference API agent-readable. Which one?

My answer — **Substitutions. Checkout is a permissions problem the industry is already standardising. Substitution is a judgment call on a live clock that only Instacart's data can answer, and today an agent can't read it at all.**

Where does an agent-run order actually get stuck? (theory of constraints)



Weighted scoring (1 to 5; weights and scores are my judgment)

What matters	Weight	Checkout	Substitutions
Needs a judgment call from the agent	25%	2	5
A person is waiting on the answer	20%	1	5
Not already solved by payment standards	15%	1	5
Needs Instacart's own data (1.6B orders)	15%	2	5
Cost of shipping it late	15%	2	4
Fits 2 engineers × 4 weeks (no PCI / fraud)	10%	3	4
Weighted total (out of 5)		**1.8**	**4.8 ✓**

My week-1 draft: one substitution rule per item in the usual basket (built from the 3 options Instacart already has)

```
{ "item": "upc:0001111041600", "mode": "specific_backups",   // or best_match / refund
  "backups": ["upc:…", "upc:…"], "max_price_change_pct": 10, "must_match": ["dairy_free"],
  "if_no_answer_after_s": 300, "fallback": "refund", "set_by": "agent:phone-assistant" }
```

Why not checkout (yet)

- Payment rails already exist: **ACP** (OpenAI + Stripe), **UCP + AP2** (Google), **Visa and Mastercard agent tokens** with spending limits [21][22].
- Checkout moved **back to the merchant**: ChatGPT's in-chat checkout was retired [12], and Claude and Gemini **hand over to Instacart** to pay [2][3].
- Real checkout work means payments, fraud and chargebacks: not a 2-person, 4-week job anyway.

Why substitutions

- Instacart's public agent tools **only create recipes and shopping lists** [9]: no way to read or set a replacement rule.
- In the aisle a person answers the shopper. **An agent can't, unless the rule is written down.**
- Instacart already has the rulebook [10] and a new **diet-aware replacement model** [6]. Expose it, don't rebuild it.

The 4 weeks, and what "done" means

- **W1** Audit what exists; design the rule format + limits
- **W2** Agents can read/set rules (API + agent tools)
- **W3** Log every decision + reason; no answer → rule, else refund
- **W4** Test: all backups out · +40% price · allergy · 2 agents clash

Done = every item in the usual basket has a readable rule, every replacement has a logged reason, all tests pass. **Would change my mind:** if agents had no way at all to pay or hand over, checkout comes first.

[2] Claude integration · [3] Gemini integration · [5] Q2 2026 results · [6] Q2 2026 call · [8] developer docs · [9] Instacart MCP docs · [10] Help Center · [12] CNBC · [21] Google Developers, UCP/AP2 · [22] PYMNTS, Visa and Mastercard agent tokens

QUESTION 4 OF 6

Two agents read “quantity: 2” differently. What went wrong?

The question: Two agent platforms read “quantity: 2” on the new API/MCP differently: one orders 2 units, one 2 × 1 kg packs of wheat. Where is the problem, and how do you fix it?

My answer — It's a contract bug, not an agent bug. A bare number plus a server that quietly picks a unit lets “2” mean two things. I'd make the unit impossible to leave out, check it against what the user expects, and have Instacart say back what it understood.

Root cause: 5 layers where “2” can lose its meaning (fishbone, most likely first)

Agent A → 2 units (any pack size) ← SAME REQUEST `{"name": "wheat flour", "quantity": 2}` → Agent B → 2 × 1 kg packs = 2 kg

Layer	What goes wrong	Likelihood	Evidence
1 API / agent contract	Bare number, unit optional: “2” has no kind (item, pack, weight).	HIGHEST	Instacart retired quantity + unit on another endpoint in 2026 [8]
2 Server resolver	Given several measures, the server quietly picks the “best” one; the agent never sees which.	HIGH	Same changelog: “Instacart can choose the best one” [8]
3 Catalog	“Wheat flour 1 kg” may be a pack, loose weight or a multipack; differs by store.	HIGH	2B+ items, 2,200 retailers [4]
4 Tool description	The tool text never says what quantity counts; each platform guesses.	MEDIUM	Public tools: create-recipe, create-shopping-list [9]
5 Agent parsing	“2 bags of atta” → data differs by model. Least controllable, so don't rely on it.	MEDIUM	My inference

✗ BEFORE: ambiguous

```
{
  "name": "wheat flour",
  "quantity": 2
}


// 2 what? item? pack?
// kg? server guesses
```

✓ AFTER: typed request

```
{ "product": {"upc": "8901725…"},
  "measure": {
    "kind": "pack", "count": 2,
    "pack_size": {"value":1,"unit":"kg"}},
  "expect_total": {"value":2,"unit":"kg"},
  "on_mismatch": "reject" }
```

🔁 READ-BACK: Instacart replies

```
{ "resolved": "2 × Atta 1 kg pack",
  "total": {"value":2.0,"unit":"kg"},
  "matched_by": "barcode",
  "check": "MATCH",
  "status": "OK" }
// shown to user + saved in log
```

Hour one: 5 Whys, starting with a diff

Replay both requests from the logs. Identical?

- YES → our side: resolver or catalog (layers 2–3)
- NO → their side: tool text or parsing (layers 4–5)

Framework I used: Poka-yoke: make the mistake impossible

1. **Typed measure:** kind is item, pack, weight or volume. The 2026 agent-protocol spec supports this kind of either/or schema [20]. **No bare quantity for agents.**
2. **Find products by barcode,** not by name; Instacart already prefers barcodes when given [8].
3. **Expected-total check:** what we resolve must equal what the agent expected, or we reject.
4. **Read-back:** Instacart replies with what it understood, so mismatches show up before the shop.
5. **Still unclear? Ask, don't guess:** the protocol can send the user a quick form [20].

Keeping it fixed

Shared test list of ~100 tricky products (multipacks, loose produce, dual units) that **every** agent partner runs before release. **Metrics:** read-back mismatch rate; a “wrong quantity” support tag. **Migration:** new format alongside the old; bare quantity rejected for agents after 90 days' notice.

[4] Clementine launch (catalog size) · [8] Instacart developer changelog and shopping-list docs · [9] Instacart MCP docs · [20] Model Context Protocol spec, 2026 (tool schemas, asking the user). Test-list size and notice period are my proposals.

QUESTION 5 OF 6

How many orders can an agent place on its own each month?

The question: Agent orders raise volume but risk "I didn't mean to order that" complaints, with no history for agent orders. What threshold per user per month, and why?

My answer — **4 orders a month on its own, with at most 1 in any rolling 7 days; the 5th asks for one tap. It covers a loyal weekly routine, so hitting it is a warning sign, and limits rise only when the evidence says so.**

4 orders a month on its own (launch limit)

max 1 per rolling 7 days · 5th+ ships after one tap · spend ≤ user's limit · opt-in only

Trust ladder: each user (and cohort) moves up only on evidence

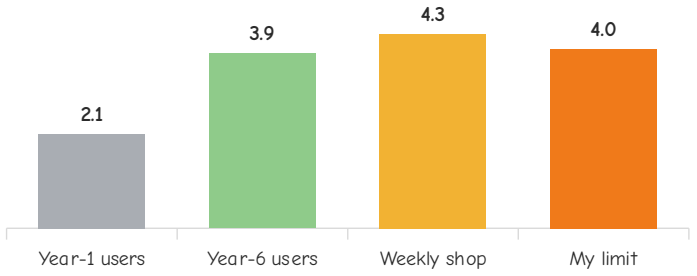
R0 DRAFT
Agent builds the basket; **you press Order.**
Learn: edit rate
Up when: opted in, 4 drafts, <10% of lines edited

R1 ORDER + HOLD
Agent orders; **60-min cancel window** before shopping starts.
Learn: cancel rate
Up when: 300 orders, 0 "didn't mean to" ⇒ <1% @95%

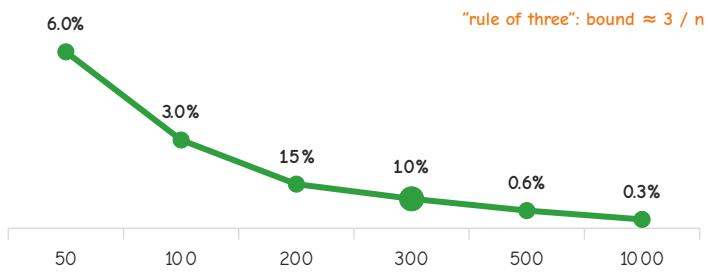
R2 ON ITS OWN
4/month, ≤1 a week, inside spend limit + substitution rules (Q3).
Learn: problems vs normal orders
Up when: 90 days at or below normal orders

R3 PERSONAL
Limit = **your own usual pace + 1** (max 8); you can raise it.
Learn: long-run trust
Tripwires can send anyone back to R0

Where 4 comes from: orders per month [7]



No history? Bound it: worst-case unwanted-order rate after n clean orders



Why 4, not 5, and not 10

- **It covers the routine.** Instacart's most loyal users average **3.9 orders/month**, year-1 users **2.1** [7]. Most people never hit 4.
- **The 5th-week tap is a feature.** Five-week months come ~4 times a year: a regular **re-consent**, the standard the FTC's 2025 order now holds Instacart to [14].
- **It limits the agent, not the customer.** Order 5 still ships after a tap, so almost no sales are lost.
- **The cost is lopsided.** One unwanted order ≈ **$115** [5] of mostly fresh food refunded or wasted. One tap ≈ 2 seconds.
- **The weekly rule catches bugs.** A monthly cap alone lets a looping agent place 4 orders in an afternoon; the weekly rule stops it at #2.
- **Not 5:** weekly users would never re-confirm. **Not 10:** 2.5× the most loyal users, so it's likelier a loop than a need.

Framework I used: Guardrail stack: every check must pass

Spend	≤ user's limit (default ≈1.25× their usual basket)
New items	never bought before → one tap
Substitutions	only inside the Q3 rules
Payment	limit enforced twice: Instacart + card token [22]

Tripwires (proposed, signed in week 2)

"Didn't mean to order" **>0.5%** of agent orders (30 days) → back to R0 · Cancels **>5%** or more problems than normal orders → freeze. I'd add a "didn't mean to order this" button so it's counted.

[5] Instacart Q2 2026 results ($115 average order) · [7] Instacart S-1 / Q3 2023 call (2017–22 cohorts) · [14] FTC v. Instacart settlement · [22] PYMNTS, Visa and Mastercard agent tokens. All thresholds are starting values to tune.

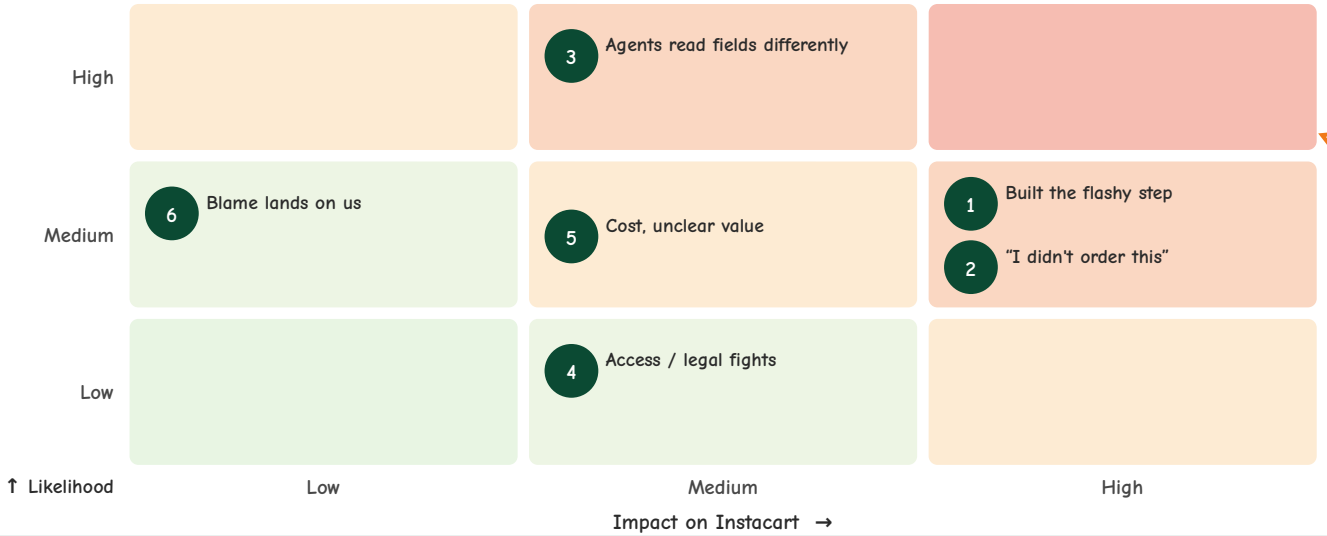
QUESTION 6 OF 6

"Should we be worried?" What I'd tell the client

The question: A competitor's agent-readiness project stalls publicly at month four: bad press, the client blames its consultants. Our Instacart contact asks: "Will this happen to us?"

My answer — **Honestly: worried enough to act, not enough to stop. Their stall tells us the category is hard, not why theirs stopped. I'd show the failure modes, what we've designed out, what we don't know yet, and the tripwires we both sign.**

Pre-mortem: how projects like this fail (risk left after our fixes)



Failure mode	Public example	What we've already designed in
1 Built the flashy step	OpenAI / Walmart in-chat checkout, pulled [11][12]	Substitutions first; demo on top (Q2, Q3)
2 "I didn't order this"	Instacart's $60M FTC case; AI price tests stopped [14][15]	Draft → hold → auto ladder; re-consent tap (Q5)
3 Fields read differently	Instacart retired its own bare quantity field [8]	Required units + read-back + test list (Q4)
4 Access / legal fights	Amazon v. Perplexity: blocked, then overturned on appeal [16]	An official front door: agent tools + login scopes
5 Cost, unclear value	Gartner: 40%+ of agent projects cancelled by 2027 [13]	4-week slices, each with a written "done"
6 Blame lands on us	The competitor's own headlines	Signed assumptions log + clear decision owners

What I'd say

"Honestly, a little, yes. Their stall tells us this is hard; it doesn't tell us why theirs stopped, and I won't guess in front of you. Here's how projects like this fail, which of those we've designed out, what we still don't know, and three points where we'd pause. If one trips, **you hear it from me first, with the data.** Let's sign them today."

Our own uncertainty, said plainly

We know

Order quality up **16 quarters in a row** [5]; AI-assistant baskets beat the $115 average [5]; live on ChatGPT, Claude, Gemini [1–3]

We don't know

How often agent orders are unwanted (no history); demand: only **22%** trust agents to buy [19]; gaps in internal APIs

When we will

Draft-order edit data after **~4 weeks**; the 300-order check for R1 around **week 12**

Three tripwires, signed in week 2

1. Agent orders have more problems than normal ones (+ agreed margin) → **freeze moving users up**
2. "Didn't mean to order" rate above the agreed level → **that group back to drafts**
3. Any quantity read-back mismatch reaching a shopper → **release blocker**

Framework I used: Pre-mortem + kill criteria + assumptions log

What I won't do: guess at their failure, distance us from the other firm, or promise zero incidents.

[1–3] ChatGPT, Claude, Gemini integrations · [5] Q2 2026 results · [8] developer changelog · [11] Search Engine Land · [12] CNBC · [13] Gartner · [14] FTC · [15] Consumer Reports · [16] Cooley, Ninth Circuit ruling · [19] FoodNavigator

Appendix

Sources. The numbers in square brackets on each slide point to this list.

1. **Instacart** · 2025 · company.instacart.com/pressreleases
   Instacart app launches in OpenAI ChatGPT, with Instant Checkout
2. **Supermarket News** · 2026 · supermarketnews.com
   Instacart connects with AI platform Claude
3. **Instacart** · 2026 · company.instacart.com/updates
   Instacart brings agentic grocery shopping to Gemini
4. **TechCrunch / PYMNTS** · 2026 · techcrunch.com
   Instacart launches AI shopping assistant Clementine
5. **Instacart** · 2026 · investors.instacart.com
   Second quarter 2026 financial results and prepared remarks
6. **Instacart** · 2026 · investors.instacart.com
   Q2 2026 earnings call: replacement model, 95% satisfaction
7. **Maplebear (Instacart)** · 2023 · sec.gov
   S-1 filing and Q3 2023 earnings call: order frequency by cohort
8. **Instacart Docs** · 2026 · docs.instacart.com/developer_platform_api
   Developer Platform changelog and create-shopping-list page
9. **Instacart Docs** · 2026 · docs.instacart.com/developer_platform_api
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    Walmart: ChatGPT checkout converted 3× worse than website
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    OpenAI revamps shopping experience in ChatGPT after Instant Checkout
13. **Gartner** · 2025 · gartner.com/en/newsroom
    Over 40% of agentic AI projects will be cancelled by end of 2027
14. **FTC** · 2025 · ftc.gov
    Instacart to pay $60 million in consumer refunds
15. **Consumer Reports** · 2025 · consumerreports.org
    Instacart stops AI pricing tests
16. **Cooley** · 2026 · cooley.com
    Ninth Circuit rules on AI agent access (Amazon v. Perplexity)
17. **Supermarket News** · 2026 · supermarketnews.com
    AI agents gaining the most trust with grocery shoppers
18. **Retail Dive / ACI Worldwide** · 2026 · retaildive.com
    Consumers trust AI to advise, but don't consent to it purchasing
19. **FoodNavigator** · 2026 · foodnavigator.com
    Grocery shoppers use AI, but trust in AI buying agents lags
20. **Model Context Protocol** · 2026 · modelcontextprotocol.io
    Specification, 2026-07-28 release: tools and elicitation
21. **Google Developers Blog** · 2026 · developers.googleblog.com
    Under the hood: Universal Commerce Protocol (with AP2)
22. **PYMNTS** · 2026 · pymnts.com
    Visa and Mastercard put tokens in charge of AI commerce
23. **FMI** · 2026 · fmi.org
    U.S. Grocery Shopper Trends 2026
24. **9to5Google** · 2026 · 9to5google.com
    Gemini Android app automation rolls out
25. **PYMNTS** · 2026 · pymnts.com
    Anthropic debuts commerce agent blueprint with Visa and Mastercard

A note on limits: everything here is public. Cohort data covers 2017–22 customers and isn't agent-specific, and surveys measure trust differently. The thresholds I propose (4 a month, 0.5%, 5 engineer-days, 300 orders) are starting points to tune, not findings.